La Scuola d'Italia / Dr. Huson / IB Math: Calculus Name:
14 April 2026

5.12 Classwork: Trigonometric Transformations & Derivatives (no calculator)

1. A Ferris wheel has a diameter of 80 metres. The lowest point of the wheel is 5 metres above the ground. The wheel completes one full revolution every 16 minutes. A seat starts at the lowest point at time $t = 0$.

   The height, h metres, of the seat above the ground after t minutes is modelled by

   $$h(t) = a\cos(bt) + d$$

   where a, b, $d \in \mathbb{R}$.

   (a) Find the value of a, b, and d. [4]

   (b) Find the height of the seat after 4 minutes. [2]

   (c) Find $h'(t)$ and state what it represents in context. [2]

2. The graph of $y = \sin x$ is transformed to give the graph of $y = 3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$.

(a) Describe the following transformations applied to $y = \sin x$:

(i) the vertical stretch, [1]

(ii) the horizontal compression, [1]

(iii) the horizontal translation, [1]

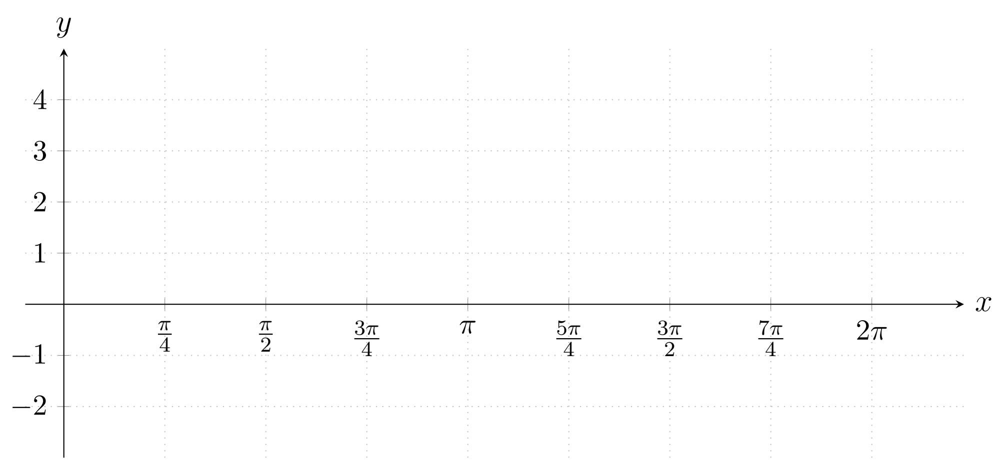
(iv) the vertical translation. [1]

(b) For the transformed function, write down:

(i) the amplitude, [1]

(ii) the period. [1]

(c) On the axes below, sketch the graph of $y = 3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$ for $0 \leq x \leq 2\pi$. Clearly label any intercepts and maximum/minimum points. [4]



3. Find the derivative of each function.

   (a) $f(x) = 3\sin x + 2\cos x$ [2]

   (b) $g(x) = 4\cos(3x)$ [2]

   (c) $h(x) = \sin^2 x$ [3]

   Hint: Write $\sin^2 x = (\sin x)^2$ *and use the chain rule.*

4. The function f is defined by $f(x) = 2\sin(3x) - 1$ for $0 \leq x \leq \pi$.

   (a) Find $f'(x)$. [2]

   (b) Find the x-coordinates of the points where f has a maximum and a minimum value. [3]

   (c) Find the maximum and minimum values of f. [2]

5. The function g is defined by $g(x) = a\sin(bx) + c$ for $0 \leq x \leq 2\pi$, where a, b, $c \in \mathbb{Z}^{+}$. The graph of g has a maximum value of 7, a minimum value of 1, and a period of $\frac{2\pi}{3}$.

   (a) Find the values of a, b, and c. [3]

   (b) Find $g'\left(\frac{\pi}{6}\right)$. [3]

6. Consider the function $f(x) = \cos 2x + 2\sin x$ for $0 \leq x \leq 2\pi$.

   (a) Show that $f'(x) = 2\cos x(1 - 2\sin x)$. [3]

   (b) Hence find the values of x for which $f'(x) = 0$. [3]